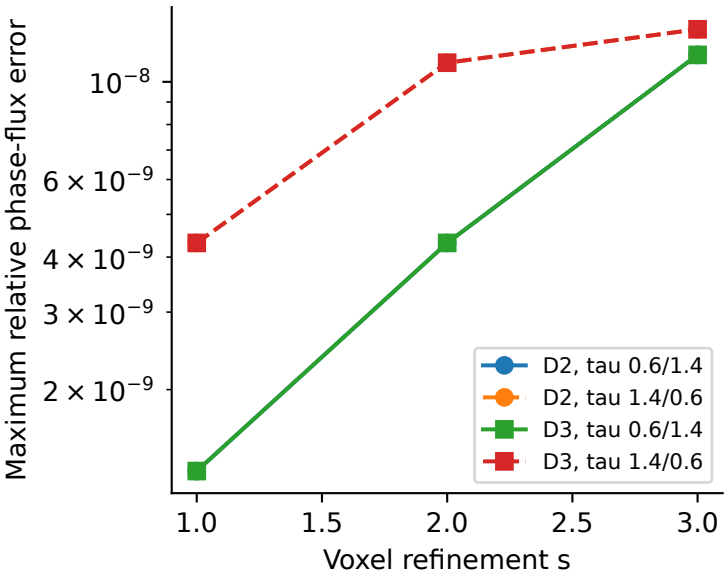
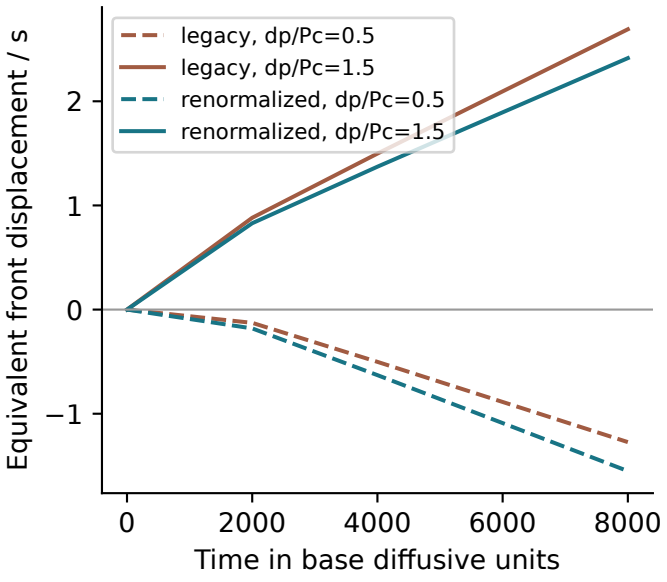
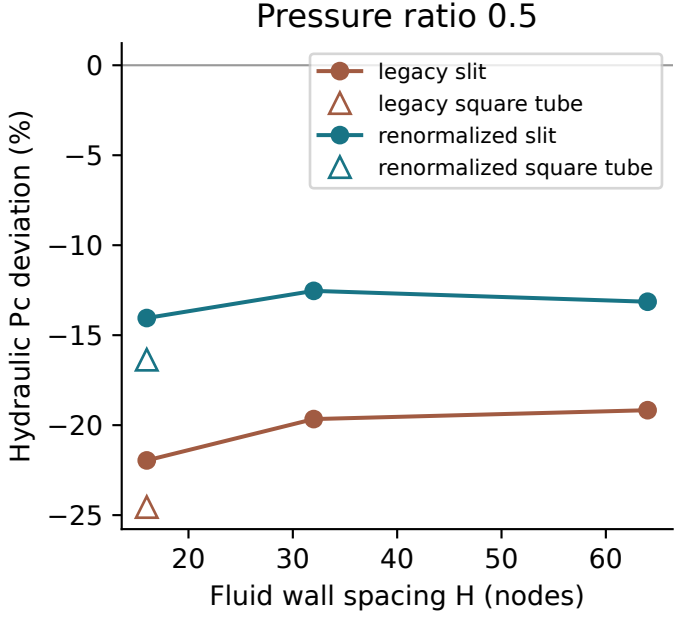
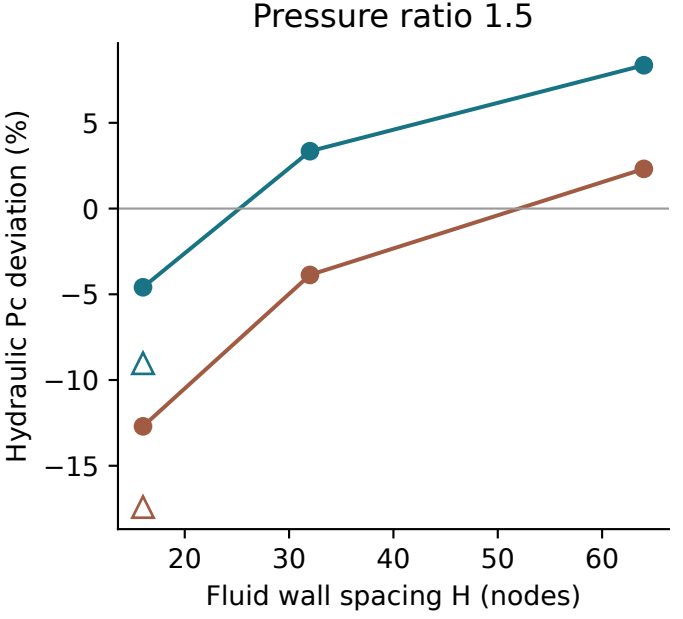

Maximum relative phase-flux error
10^{-8}
6×10^{-9}
4×10^{-9}
3×10^{-9}
2×10^{-9}
D2, tau 0.6/1.4
D2, tau 1.4/0.6
D3, tau 0.6/1.4
D3, tau 1.4/0.6
1.0
1.5
2.0
2.5
3.0
Voxel refinement s


Equivalent front displacement / s
legacy, dp/Pc=0.5
legacy, dp/Pc=1.5
renormalized, dp/Pc=0.5
renormalized, dp/Pc=1.5
2
1
0
−1
0
2000
4000
6000
8000
Time in base diffusive units


Pressure ratio 0.5
Hydraulic Pc deviation (%)
legacy slit
legacy square tube
renormalized slit
renormalized square tube
0
−5
−10
−15
−20
−25
20
30
40
50
60
Fluid wall spacing H (nodes)


Pressure ratio 1.5
Hydraulic Pc deviation (%)
5
0
−5
−10
−15
20
30
40
50
60
Fluid wall spacing H (nodes)